

Xun Gong

FINAL-YEAR PH.D. CANDIDATE AT SJTU

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About myself

I am a **final-year** Ph.D. candidate at the School of Computer Science, Shanghai Jiao Tong University, advised by Prof. Yanmin Qian. My research focuses on speech understanding, particularly end-to-end unified modeling for large speech-language models (Speech LLMs), as well as automatic speech recognition (ASR) and speaker-related tasks.

In the area of Speech LLMs, I work on **cross-modal alignment** between speech and text, **post-training** techniques, and **fine-grained context modeling**. I am also involved in designing unified architectures for full-duplex speech interaction. My work on ASR emphasizes adaptation to **dialects, domains, and hotwords**. Additionally, I study speaker verification and target speaker ASR to further enhance robustness and personalization in speech systems.

I earned my B.S. in Computer Science from the ACM Honor Class of Zhiyuan College, SJTU, where I was advised by Prof. Yong Yu.

Education

Ph.D. Candidate of Computer Science

ADVISED BY PROF. **YANMIN QIAN**

Sept. 2021 - June 2026

Shanghai Jiaotong University, China

- Member of Auditory Cognition and Computational Acoustics Lab (AudioCC Lab) in Shanghai Jiaotong University.

Bachelor of Science (B.S.) in Computer Science

ADVISED BY PROF. **YONG YU**, PROF. **WEINAN ZHANG**

Sept. 2017 - June 2021

Shanghai Jiaotong University, China

- Member of ACM Honor Class, which is an elite CS program for top-5% talented students.
- Average Score of Professional Courses: 93/100

Internship Experience

Research Intern

Advised by Prof. **Yanmin Qian**

Jan. 2025 – Oct. 2025

AudioCC Lab / Yusheng Yueban,
Shanghai, China

Research Intern

Advised by Dr. **Zhimin Wang**

Jan. 2024 – Dec. 2024

Ant Group, Shanghai, China

Research Intern

Advised by Partner Applied Science Manager **Jinyu Li** and Principal Research Manager **Shujie Liu**

July 2022 – Jan. 2023

Microsoft Research Asia, Beijing, China

Undergraduate Research Intern

Advised by Dr. **Tian Tan** and Prof. **Yanmin Qian**

May 2021 – Dec. 2021

AlSpeech, Suzhou, China

Undergraduate Algorithm Intern

Advised by Prof. **Yanmin Qian** and Prof. Kai Yu

Apr. 2019 – Nov. 2020

X-LANCE, Shanghai Jiao Tong University,
Shanghai, China

Research Contributions

“Jiaojiao” : Academic In-House Speech Interaction Large Model (Voice Interaction SpeechLLM)

Apr. 2025

Jiaojiao is a **GPT-4o-like** end-to-end speech large language model (**Speech LLM**) supporting multilingual understanding, real-time voice cloning, role playing, and emotional expression. It covers Mandarin dialects, multilingual multi-speaker, and multi-scenario dialogue capabilities. We designed and implemented a dual-modality understanding-generation architecture, achieving strong performance on semantic understanding and speech recognition benchmarks, and promoted it to become the core module of my advisor's startup project.

AudioCC Lab / Yusheng Yueban,
Shanghai, China

Contextual Enhanced SpeechLLM with Large-Scale Hotword Biasing

Sept. 2024

Modern SpeechLLMs impose higher demands for accurate recognition of hotwords (bias words such as person names, place names, and rare terms). We propose the **Contextual-Bias** SpeechLLM and a Bias Retrieval framework. On LibriSpeech test sets, we support up to **200K bias words**, achieving significantly better performance than all existing systems while maintaining low latency. Additionally, we developed and solidified the codebase for fine-tuning speech large models and successfully applied the system in real-world deployment.

Ant Group, Shanghai, China

Long-Content Enhanced Speech Recognition (Long-History Helps ASR)

Jan. 2023

For **long-form conversational scenarios**, we propose new LongFNT and SLongFNT architectures that leverage both textual and speech history for modeling, aimed at improving ASR performance. On LibriSpeech and GigaSpeech, our system achieves up to 26% relative WER reduction. The results were published in the journal TASLP.

Microsoft Research Asia, Beijing, China

Text-Only Domain Adaptation in ASR

Jan. 2024

To address text-only domain adaptation, we developed a three-stage end-to-end adaptation framework: designed a plug-and-play language model fusion mechanism, synthesized diverse speech via splice-TTS for fine-tuning, and proposed a domain-discriminative decoding strategy to improve inference robustness. On the LibriSpeech-to-GigaSpeech transfer task, we achieved up to 40% WER improvement over conventional methods.

AudioCC Lab, Shanghai, China

Accent Adaptation in ASR

Nov. 2021

Proposed an end-to-end adaptation architecture for multi-accent speech recognition, combining PPG features and TTS-based augmentation to obtain accurate accent representations. For recognition, designed gated and multi-basis adaptation layers, significantly improving ASR performance across various accents. During the internship, I also contributed to speaker verification, identification, and separation tasks, developing multiple recognition models based on Kaldi.

AISpeech, Suzhou, China

Awards and Honors

SJTU Graduate Scholarship

2022–2025

Wu Wenjun Artificial Intelligence Scholarship

2022–2025

Zhiyuan Honor Scholarship

2017–2021

Outstanding Undergraduate Graduate Award, Shanghai Jiao Tong University

2021

Publications

“Jiaojiao” : A Groundbreaking Spoken Dialogue Emotion Large Model Rivals GPT-4o Developed

Apr. 2025

Xun Gong, Wei Wang, Bing Han, Leyin Zhang, Haiyang Sun, Chenyang Le, Chenda Li, Yanmin Qian

InterSpeech 2025 Festival

BR-ASR: Efficient and Scalable Bias Retrieval Framework for Contextual Biasing ASR in Speech LLM

Jan. 2025

Xun Gong, Anqi Lv, Zhiming Wang, Yanmin Qian

InterSpeech 2025

Ranking and Selection of Bias Words for Contextual Bias Speech Recognition

Jan. 2025

Haoxiang Hou, Xun Gong, Wangyou Zhang, Yanmin Qian

InterSpeech 2025

Integrative Strategies for Text-Only Domain Adaptation in End-to-End Speech Recognition

Jan. 2025

Xun Gong, Wei Wang, Hang Shao, Yanmin Qian

Submitted to TASLP

Contextual Biasing Speech Recognition in Speech-enhanced Large Language Model

Sept. 2024

Xun Gong, Anqi Lv, Zhiming Wang, Yanmin Qian

InterSpeech 2024

Whisper-KDQ: A Lightweight Whisper via Guided Knowledge Distillation and Quantization for Efficient ASR

Oct. 2024

Hang Shao, Wei Wang, Bei Liu, Xun Gong, Haoyu Wang, Yanmin Qian

SLT 2024

Advanced Long-Content Speech Recognition with Factorized Neural Transducer

Dec. 2023

Xun Gong, Yu Wu, Jinyu Li, Shujie Liu, Rui Zhao, Xie Chen, Yanmin Qian

TASLP

Text Only Domain Adaptation with Phoneme Guided Data Splicing for End-to-End Speech Recognition

Aug. 2023

Wei Wang, Xun Gong, Hang Shao, Yanmin Qian

InterSpeech 2023

Joint Discriminator and Transfer Based Fast Domain Adaptation For End-To-End Speech Recognition

Hang Shao, Tian Tan, Wei Wang, **Xun Gong**, Yanmin Qian

[Jun. 2023](#)

ICASSP 2023

Factorized AED: Factorized Attention-Based Encoder-Decoder for Text-Only Domain Adaptive ASR

Xun Gong, Wei Wang, Hang Shao, Xie Chen, Yanmin Qian

[Jun. 2023](#)

ICASSP 2023

LongFNT: Long-Form Speech Recognition with Factorized Neural Transducer

Xun Gong, Yu Wu, Jinyu Li, Shujie Liu, Rui Zhao, Xie Chen, Yanmin Qian

[Apr. 2023](#)

ICASSP 2023

Speechlm: Enhanced speech pre-training with unpaired textual data

Ziqiang Zhang, Sanyuan Chen, Long Zhou, Yu Wu, **Xun Gong**, and others

[Dec. 2022](#)

arXiv

Knowledge Transfer and Distillation from Autoregressive to Non-Autoregressive Speech Recognition

Xun Gong, Zhikai Zhou, Yanmin Qian

[Sept. 2022](#)

Interspeech 2022

Layer-wise Fast Adaptation for End-to-End Multi-Accent Speech Recognition

Yanmin Qian, **Xun Gong**, Houjun Huang

[Aug. 2022](#)

TASLP

Knowledge Transfer and Distillation from Autoregressive to Non-Autoregressive Speech Recognition

Xun Gong, Zhikai Zhou, Yanmin Qian

[Apr. 2022](#)

Interspeech 2022

The SJTU System For Multimodal Information Based Speech Processing Challenge 2021

Wei Wang, **Xun Gong**, Yifei Wu, Zhikai Zhou, Chenda Li, Wangyou Zhang, Bing Han, Yanmin Qian

[May 2021](#)

ICASSP 2022

Layer-wise Fast Adaptation for End-to-End Multi-Accent Speech Recognition

Xun Gong, Yizhou Lu, Zhikai Zhou, Yanmin Qian

[Jan. 2021](#)

Interspeech 2021

Speaker Embedding Augmentation with Noise Distribution Matching

Xun Gong, Zhengyang Chen, Shuai Wang, Yanmin Qian

[Jan. 2021](#)

ISCSLP 2021

Text Adaptation for Speaker Verification with Speaker-Text Factorized Embeddings

Yexin Yang, Shuai Wang, **Xun Gong**, Yanmin Qian, Kai Yu

[Oct. 2019](#)

ICASSP 2020

End-to-End Texture-Aware and Depth-Aware Embedded Advertising for Videos

Xun Gong (Equal Contribution), Jiasen Li, Boning Li

[Nov. 2019](#)

ICCTA 2020

Patents

Hotword Retrieval in Speech Multimodal Large Models

Xun Gong, Anqi Lv, Zhiming Wang

[May 2025](#)

P250511CN

Hotword Recognition in ASR Based on Speech Multimodal Large Models

Xun Gong, Anqi Lv, Zhiming Wang

[Sep. 2024](#)

CP250259MY

Compression of Pretrained Models Based on Knowledge Distillation and Quantization

Yanmin Qian, **Xun Gong**

[Aug. 2023](#)

CN116629328A

Phoneme-Based Transfer Learning Method for Speech Domain

Yanmin Qian, **Xun Gong**

[Apr. 2023](#)

CN115985290A

Knowledge Distillation Method, Electronic Device, and Storage Medium

Yanmin Qian, **Xun Gong**, Tian Tan

[Jul. 2022](#)

CN114822518A

Method and Apparatus for Multi-Accent Speech Recognition

Yanmin Qian, **Xun Gong**

[Nov. 2021](#)

CN113593534A

Speaker Verification Method and Apparatus

Yanmin Qian, **Xun Gong**, Shuai Wang

[Apr. 2020](#)

CN111081255A

Speech Data Augmentation Method and Apparatus

Yanmin Qian, **Xun Gong**

[Oct. 2020](#)

CN111863003A